



Using deep learning predictions to study the development of drawing behaviour in children^{*}

Benjamin Beltzung^a, Marie Pelé^b, Lison Martinet^a, Elliot Maitre^c, Jimmy Falck^d,
Cédric Sueur^{a,e,*} 

^a Université de Strasbourg, CNRS, IPHC UMR 7178, Strasbourg, France

^b ANTHROPO-LAB – ETHICS EA 7446, Université Catholique de Lille F-59000 Lille, France

^c Université de Toulouse - IRIT UMR5505, 31400 Toulouse, France

^d Laboratoire lorrain de recherche en informatique et ses applications (Loria – CNRS/Université de Lorraine/Inria), Nancy, France

^e Institut Universitaire de France, Paris, France

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ABSTRACT

Drawing behaviour in children provides a unique window into their cognitive development. This study uses Convolutional Neural Networks (CNNs) to examine cognitive development in children's drawing behaviour by analysing 386 drawings from 193 participants, comprising 150 children aged 2–10 years and 43 adults from France. CNN models, enhanced by Bayesian optimization, were trained to categorize drawings into ten age groups and to compare children's drawings with adults' ones. Results showed that model accuracy increases with the child's age, reflecting improvement in drawing skills. Techniques like Grad-CAM and Captum offered insights into key features recognized by CNNs, illustrating the potential of deep learning in evaluating developmental milestones, with significant implications for educational psychology and developmental diagnostics.

1. Introduction

In children, drawing behaviour appears around the age of 18 months-old. During a lifetime, drawing is an important mode of communication driven not only by cognitive aspects but also by cultural ones. For instance, by utilising human figure representations in drawings [1], demonstrated variations in body size and shape among young adults from Israel and Thailand when asked to depict themselves. Similarly, through analysing drawings created by children from different cultures, Restoy and colleagues [2] showcased how the level of individualism in countries could influence the size and number of human figures depicted. Of course, culture is not the sole factor influencing drawing behaviour and numerous cognitive aspects have been explored through it. Luquet [3] firstly suggested that the development of the drawing behaviour can be seen as a four-stage process. The first stage occurs when the child lacks intention to represent reality and discovers by chance a shape analogy between an object and its initially nonsignificant trace (*fortuitous realism*). Around four or five years old, the child tries to produce realistic drawings but does not have the entire abilities

(motor and representative skills) to do so (*missed realism*). Then, the child uses her knowledge of objects' components to represent them but some misconceptions linked for example to transparency or perspective representation are still present (*intellectual realism*). Eventually, the last stage proposed by Luquet [3] is *visual realism*, when every step is completed.

Since, many other developmental theories on drawing behaviour have been proposed. For example, Adi-Japha and colleagues [4] proposed three different steps in the development of drawing behaviour in children. First, *action representation* occurs when the drawing is associated with verbalisation. For example, when the child represents a moving object like a car or a train and produces the corresponding sound. *Romancing* occurs when the child is able to name her drawing but it remains challenging for another individual to interpret it. Then, the *guided elicitation* phase occurs when the child is able to produce a figurative drawing helped by an adult. Thus, Adi-Japha and collaborators [4] focused on *how* the drawing is produced better than the drawing as a result. On the other hand, Baldy [5] proposed a classification only based on the morphological development of the human figures in the drawing

^{*} This paper was recommended for publication by Prof Guangtao Zhai.

^{*} Corresponding author at: IPHC UMR 7178, 23 rue Becquerel, 67087 Strasbourg, France.

E-mail address: cedric.sueur@iphc.cnrs.fr (C. Sueur).

(e.g. tadpoles, filiform figures, tube-shaped, etc.). Doing so, Baldy's classification appears more focused on the drawing as a product and then, is closer to Luquet's classification.

Even if all these developmental theories are relevant and not intrinsically exclusive, they consider different concepts and focused on different aspects of the drawing behaviour: its process and its result. Without forgetting that it may be difficult to visually interpret the corresponding stage for a given drawing. Traditionally, drawings are analysed by defining and extracting a set of features such as the number and size of figures or the number of used colours, etc. While this approach can be insightful, two challenges arise. First, the amount of information contained in a drawing is substantial in nature, and using predefined features significantly limits the amount of extracted information. Then, as each of these features focuses on a single aspect of the drawing, this approach does not consider the holistic aspect of drawings (i.e. considering drawing as a whole, and not only its different parts). Moreover, studies in toddlers showed that their drawings may have some meanings even if they do not have figurative aspects on it [6,7]. For these reasons, such methods may not be sufficient to benefit from the information contained in a drawing to its fullest extent. A possible way to mitigate these issues is to ask children about what they intended to represent. However, they may not be directly conscious about the deep meaning of their drawing, and this could not be applied to scribbles drawn by very young children who are not able to verbally communicate yet or children with pathologies that make them unable to communicate.

To minimise these biases, a potential candidate is the use of artificial intelligence and more precisely deep learning. Over the past decade, important advances in deep learning models (neural networks) have been made when considering images, video or audio processing, substantially improving the predictive accuracy and outperforming state-of-the-art methods in many fields, such as system health management [8], face recognition [9], or even speech recognition [10]. The most popular type of deep learning models is Convolutional Neural Networks (CNNs) [11], which is known to provide a high accuracy for tasks involving image analyses. While CNNs architecture can vary according to the task, some key concepts remain. Different types of layers exist in CNN and play different roles. In convolutional layers, a convolution is applied on the image: a filter, representing a feature, slides over the image, and results in a feature map. Each value of the feature map is the degree of activation of the filter on the corresponding part of the image. Depending on the depth of the convolutional layers, filters can detect either low-level features (e.g. lines or curves) or high-level features (e.g. objects).

While such models usually provide a high accuracy, they are widely considered as 'black boxes' with regard to the decision-making process [12,13], as it is not possible to directly understand the process that led the model to predict a particular output. Indeed, the complexity and number of parameters that can attain billion, make such models difficult to interpret. However, from this complexity also comes a strength. Deep learning models allow for analysing every pixel of a given image, extracting a large amount of information contained in it, and can therefore potentially grasp all the relevant features. The features learning produced by neural networks are also complex, and allow for an objective feature representation.

Although deep learning has already been successfully used to analyse drawings [14–17], even in children [18,19]; the development of the drawing behaviour has been, to our knowledge, only poorly analysed through the lens of deep learning [20–23].

In this paper, we first build and train a CNN by using Bayesian optimisation to classify drawings according to the age of the individuals (i.e. 10 age categories, from children to adults). By using the same method, we then trained multiple models to classify children sub-categories versus adults' drawings to compute the accuracy and predict drawings not belonging to the classes considered in the models. We hypothesised that the accuracy of the models should increase with the age difference between children and adults. Indeed, as children grow,

their drawing skills improve and their drawings become closer to what an adult could produce.

2. Material and methods

2.1. Dataset

The dataset consists of 386 drawings produced by 150 children (from two to ten years-old) and 43 adults (novice and experts, see below) (details in Table S1). All adult participants were French, while all children were either French or had been living in France for many years, with all children attending the same school. In France, school classes are typically organized based on the calendar year of birth, meaning children of the same birth year are grouped together. Therefore, in our study, age categories correspond to children within the same birth year. All the participants were given a touch-screen (iPad) displaying a white background and could choose among a panel of ten colours by having access to an overlay on the bottom of the screen.

From three years-old to adults, the drawings were collected using the following protocol. Each participant was asked to produce two drawings, one under a free condition, where the participant was not given a particular instruction, and one under a self-portrait condition, where the drawer was asked to draw him- or herself [24]. The rationale behind selecting one free drawing and one self-portrait lies in balancing controlled conditions with exploratory diversity [24]. Self-portraits provide a standardized task often used in psychology to study the development of representing human figures, allowing for more structured comparisons. On the other hand, free drawings encourage creative expression, enabling the inclusion of diverse elements that reveal a broader range of differences across participants. Additionally, younger children may struggle to comprehend the instructions for a self-portrait, making free drawing a more accessible option for capturing their developmental stage and artistic capabilities.

As two-year-old participants were not verbally able to communicate and to understand the instructions, no particular task was given for these productions. For this reason, they will be considered as produced in free condition. In this category, 6 subjects participated and produced a total number of 30 drawings, ranging from 3 to 8 drawings per individual (respectively 3, 4, 4, 5, 6, 8 drawings).

Concerning adults, two categories were defined prior to the data collection. First, a group of adults who had never taken drawing classes and did not have drawing as a hobby, that will be here considered as novices. Then, an expert category, including art school students and professional illustrators.

All drawings were of dimensions 2732×2048 , and were resized to a 224×224 square with 3 channels for the colours to conduct analyses. Examples of drawings are shown in Fig. 1. The data distribution is presented in Table S1. For more information about these datasets, please refer to [24].

2.2. Ethics

We ensured the confidentiality of drawings collected from human participants, adhering strictly to the ethical guidelines of our research institutions. The study received approval from the Strasbourg University Research Ethics Committee (Unistra/CER/2019–11). Informed consent was secured from all adult participants and from a parent or legal guardian for minors. Additionally, consent for the publication of any identifying images in an online open-access format was obtained, safeguarding participant privacy and adhering to ethical standards for research.

a. Transfer learning from deep learning.

To analyse the development of the drawing behaviour through the age, we compared drawings from different age categories. To do so, we used deep learning, and more precisely transfer learning. Transfer learning is a method used in machine learning and consists in using the



Fig. 1. Examples of drawings illustrating the variability of production across different age groups and drawing conditions. The figure is structured as a 2×3 grid, where the columns represent three distinct age categories: (a) 3-year-old children, (b) 7-year-old children, and (c) expert adults, and the rows represent two experimental conditions. The first row (top) shows drawings produced in the free-drawing condition, where participants were invited to draw anything they wished, allowing spontaneous creativity and imagination to emerge. The second row (bottom) shows drawings from the self-portrait condition, where participants were explicitly asked to draw themselves, reflecting self-perception, body awareness, and the ability to represent human facial or bodily features. This arrangement provides a visual comparison of developmental progression in drawing abilities across age and task, from the exploratory and often abstract productions of young children, through the increasingly structured and representational drawings of older children, to the refined and technically skilled works of expert adults. Notably, the figure highlights contrasts both within age groups across tasks (free vs. self-portrait) and across age groups within tasks, offering insights into how drawing style, representational strategies, and graphic conventions evolve with age and cognitive development.

knowledge of an already trained model for another task [25]. This technique is particularly useful for small datasets, which is the case here. For this reason, we also used data augmentation, more particularly horizontal flips, as this transformation does not distort the image and the result remains realistic. For this study, we used the architecture of VGG19 [26] pretrained on the ImageNet dataset [27], as VGG models have already been widely used for drawing analyses [20,28]. VGG19 is a CNN consisting of 16 convolutional layers and 3 fully connected layers. The last fully connected layer was removed, as the ImageNet classes are not of interest in this study.

By using VGG19 architecture and ImageNet weights, we first trained a model with 10 classes (i.e. the age categories). The architecture of all models is described below. Then, to refine these analyses, we considered, for children only, new age categories by grouping each class with the following one. For example, drawings produced by 3 and 4 years-old are gathered in a new category called '3–4 years-old', drawings produced by 4 and 5 years-old will be in a new category called '4–5 years-old', thus approximately doubling the number of drawings per category. This process was done up to and including 10 years-old. Novices and experts' drawings were independently grouped in a new category simply called 'adults', in order to compare drawings from children to those of adults. To do so, we trained 7 models, each classifying drawing from a following pair of children's categories against the adults' drawings. A unified 10-class CNN was first trained (accuracy $\approx 40\%$; see §3.1). Because its confusion matrix showed substantial overlap between neighbouring child ages, we next trained seven binary child-vs-adult models (2–3 yrs ... 8–9 yrs). These small, balanced models sharpen age-specific discrimination and, when their calibrated probabilities are aggregated (§2.6), yield a continuous maturity index that the single multi-class network could not provide.

We did exactly the same protocol with ResNet18 model [29]. ResNet18 is a model from the Residual Network (ResNet) family consisting of 18 layers. We obtained results with ResNet18 similar to the ones of VGG19. Codes and scripts are available on Github: <https://github.com/cedricsueur/drawinganalyses>.

The convolutional neural networks (CNNs) used in our study, including VGG19 and ResNet18, extract specific features that represent

the underlying developmental and structural traits in the drawings. These features include spatial organization, shape complexity, proportional relationships, and colour or texture contrasts, which are crucial for distinguishing between age categories and between children's and adults' drawings. By leveraging the hierarchical structure of these architectures, the models systematically capture low-level features (e.g., edges and curves) and progressively build higher-level representations (e.g., shapes, patterns, and structural arrangements). The advantage of using these pre-trained models lies in their ability to generalize effectively across tasks while enabling automated refinement of the architecture through systematic optimization. Compared to traditional methods, this approach offers scalability, objectivity, and the ability to adapt dynamically to different datasets. Additionally, tools like Grad-CAM (Gradient-weighted Class Activation Mapping) and Captum provide insights into the decision-making processes of these models, enhancing interpretability by visualizing the regions of drawings that contribute most to classifications. The use of both VGG19 and ResNet18 allowed us to validate the robustness of the extracted features and demonstrate the utility of an ASD framework for drawing analysis, particularly in capturing developmental and stylistic nuances.

2.3. Bayesian optimization

For the training, we independently built models and tuned hyperparameters. Multiple strategies exist to find the best architecture and hyperparameters. For example, grid search consists in defining a set of vectors for each hyperparameter and training a model for every possible combination. Another possibility is a random search, taking into account the fact that hyperparameters may not all have the same impact.

Here, we used Bayesian optimisation, a technique using the information of past evaluations to iteratively find the best combination from a given parameters space. The model is first trained with a set of parameters subjectively defined by the user. This method then allows for using prior knowledge (i.e. the knowledge of the past trials) to select a new set of parameters that are expected to improve the accuracy. The number of iterations of this process is defined beforehand.

The architecture of our model is based on VGG19, which takes an

image with a (224,224,3) shape as input. Before flattening the result of the last convolution, we considered 3 pooling options through the Bayesian optimisation process: average pooling, max pooling, or no pooling. After flattening, we considered a dropout layer through Bayesian optimisation, with a value between 0 and 0.4 and a step (i.e. the smallest meaningful distance between two values) of 0.1, potentially followed by a Batch Normalisation layer. Then, from 1 to 3 fully connected layers are added, each containing from 32 to 512 units with a step of 32, each followed by a ReLU activation. The last fully connected layer is eventually followed by the classification layer using sigmoid function and binary cross-entropy loss. The learning rate is also optimised between $1e^{-5}$ and $1e^{-1}$ through log sampling with stochastic gradient descent (SGD) optimiser. The tuner is running on 50 trials (i.e. testing 50 different combinations of parameters) to minimise the validation loss. For each trial, the model is trained on 15 epochs, with a batch size varying from 16 to 64 with a step of 16.80 % of the data are used for the training and 20 % for validation, with an early stopping if the validation accuracy did not improve for the last 3 epochs. Once all the trials have been evaluated, the optimal parameters are saved. The best architecture with the best hyperparameters is then trained 10 times to compute the validation accuracy and finally the mean validation accuracy for each model. To assess the relative importance of colours, the same procedure was conducted after converting the images into grayscale. For a given age category, the same hyperparameters determined through Bayesian optimization were used across models, with random seeds ensuring random initialization of the layers specific to the model. Additionally, horizontal flips were applied as a data augmentation technique, with images selected randomly for flipping, introducing variability between training iterations.

2.4. Predictions matrices

To clarify how binary classifiers were utilized to build a multi-class (10-class) model, we leveraged the architecture of our models, which were based on VGG19, to systematically extend binary classification into a multi-class framework. Each model utilized the sigmoid function at its classification layer and was trained using binary cross-entropy loss, enabling precise differentiation between two classes, such as specific age categories or between children and adults. In our approach, each binary model focused on one pair of classes (e.g., children aged 3–4 years vs. adults) and optimized its parameters through Bayesian optimization. The binary cross-entropy loss function was integral to this process, as it minimized the discrepancy between the predicted probabilities (from the sigmoid function) and the true binary labels (0 for children, 1 for adults). Specifically, we trained seven independent binary classifiers, each contrasting one successive one-year child cohort with the pooled adult drawings: 2–3 yrs, 3–4 yrs, 4–5 yrs, 5–6 yrs, 6–7 yrs, 7–8 yrs, and 8–9 yrs vs Adults. Adults were used as a developmentally stable reference—their drawings show far lower intra-group variability—providing a robust anchor for positioning every child cohort on the maturity continuum reflected by the model probabilities (0 = child-like, 1 = adult-like). This ensured the model's robustness in accurately distinguishing between its designated classes. To extend this binary classification into a multi-class perspective, we employed prediction matrices to aggregate the probability outputs from multiple binary models. Each matrix captured the average prediction scores across different models, offering insights into age-group similarities and distinctions. For instance, a model trained to differentiate between 3 and 4-year-olds and adults could provide meaningful predictions for a drawing by a 10-year-old child, as the sigmoid-generated probability scores could be analysed in the context of other binary models. This matrix-based integration allowed us to infer the broader class membership and enabled a nuanced understanding of drawing characteristics across all 10 age groups. The use of binary cross-entropy loss throughout the architecture was crucial in ensuring the interpretive reliability of these models. By focusing on pairwise classifications and aggregating results

across multiple binary models, we effectively constructed a framework that combined the strengths of binary classification with the flexibility and interpretive depth required for multi-class analysis. This methodology allowed us to comprehensively analyse the development of drawing behaviour across a wide spectrum of ages while leveraging the robustness of the binary cross-entropy approach. Although every binary model shares the same VGG-19 architecture and is trained on balanced (50 % child/50 % adult) data — so that a raw sigmoid output of 0.5 always marks chance performance — raw probabilities can still be miscalibrated across models. To ensure that the values aggregated in the prediction matrices are directly comparable, we therefore applied a two-step procedure: 1. Temperature scaling was fitted on a common held-out validation set (10 % of the full dataset) for each classifier, following the reliability-curve approach of Guo et al. (2017, [30]). This brought every model's Expected Calibration Error below 0.02. 2. The calibrated probabilities p_{pp} were converted to log-odds $\log(p/(1 - p))$ and z-standardised with respect to their own validation-set distribution. The resulting values were finally back-transformed into the [0,1] interval so that a score of 0 corresponds to drawings that are maximally child-like, 1 to drawings that are maximally adult-like, and intermediate scores quantify the relative “maturity” of a drawing on a common, model-independent scale. Consequently, a score of 0.6 now conveys the same evidence for “adult-likeness” whichever binary model produced it, allowing us to compare and average predictions across the seven child-vs-adult classifiers with full statistical validity.

To evaluate the effects of the age categories and colours (RGB or grayscale) on accuracy, we performed a two-way analysis of variance (ANOVA) with interaction. Before conducting the ANOVA, we checked the assumptions of normality and homogeneity of variance. The Shapiro-Wilk test was applied to the residuals of the model to test for normality ($W = 0.98479$, $p = 0.1239$), confirming that the residuals followed a normal distribution. Homogeneity of variance was assessed using Levene's test ($F = 0.9943$, $p = 0.4609$), which indicated equal variances across groups. The ANOVA model evaluated the main effects of categories and colours, as well as their interaction, on accuracy. Post-hoc comparisons were performed using Tukey's HSD test to identify significant differences between specific group combinations. All statistical analyses were conducted in R.

2.5. Explicability

In our method, we also employed Captum, a model interpretability library for PyTorch, alongside Grad-CAM for explicability purposes. Deep learning models are known to be very efficient for image classification, however, most models remain as ‘black boxes’, and disentangling features which played a role in the classification remains a challenging task. This complexity arises from the difficulty in interpreting such models, as highlighted in recent studies [15]. In the present case, it is of interest to understand which features were discriminative. However, given that a significant proportion of drawings are nonfigurative, directly answering this question proves to be challenging. Instead, a viable approach involves examining the regions of the images that played an important role in the classification.

To this end, Grad-CAM [31] offers a powerful method. For any given image, Grad-CAM generates a heatmap that highlights the regions important for a specific class. This becomes particularly insightful when applied to the predicted class to discern why the model categorised the input image as belonging to this class. The algorithm computes the gradient using the activation of the last convolutional layer, which captures high-level features, for a given class.

Integrating Captum [32] into this workflow enhances the explicability further by providing a comprehensive toolkit for model interpretability. Captum supports various interpretability algorithms, including Grad-CAM, allowing researchers to not only visualise important regions but also understands the attribution of each input features to the model's output. By applying Captum's Grad-CAM visualisations

on the validation data for every model, we can gain deeper insights into the discriminative features recognised by the models. This integration facilitates a more nuanced understanding of model predictions, particularly in complex cases where direct interpretation of features is not straightforward. Feature maps generated by Grad-CAM and Captum were systematically analysed to identify the regions the model deemed significant for classification. These maps revealed varying focus on elements such as faces, spatial organization, and non-painted areas, providing insights into the model's feature extraction process.

3. Results

3.1. 10-classes model

The optimal model for classifying across drawing conditions achieved an accuracy of 88 %. A theoretical accuracy of 66.1 % was calculated by combining the probabilities of correct classifications for each drawing condition, based on the proportion of drawings in each category (free drawing and self-portrait) and the frequency of faces appearing in those categories (40.5 % in free drawing and 91.7 % in self-portrait). This is interesting because 88 % is much higher than 66.1 % meaning that the optimal model bases its condition recognition not only

on faces but also on other parameters.

The optimal model for classifying drawings across ten age categories achieved the accuracy of 40 %, as illustrated in the confusion matrix presented in Fig. 2. This performance significantly surpasses the 10 % accuracy expected from a model making predictions at random. Although the presence of a diagonal in the confusion matrix indicates correct classifications, a considerable number of drawings were incorrectly predicted by the model to belong to ages other than their true categories. To enhance our understanding of the model's performance and address its limitations, we went further into the analysis of the model. This involved examining the patterns and characteristics of the misclassifications to identify potential areas for improvement and gain insights into the model's decision-making process.

To gain deeper insights into the distinctions between our age groups, we trained models to differentiate between drawings made by adults and those made by children of various age categories. The selection of the optimal models was facilitated through Bayesian optimisation, with the chosen parameters and their mean accuracy detailed in Table S2. This mean accuracy was derived from the validation accuracy of the optimal model, which was trained ten times. The performance of models trained on grayscale drawings is specifically outlined in Table S3. An interesting trend observed is the decrease in model accuracy with increasing age for

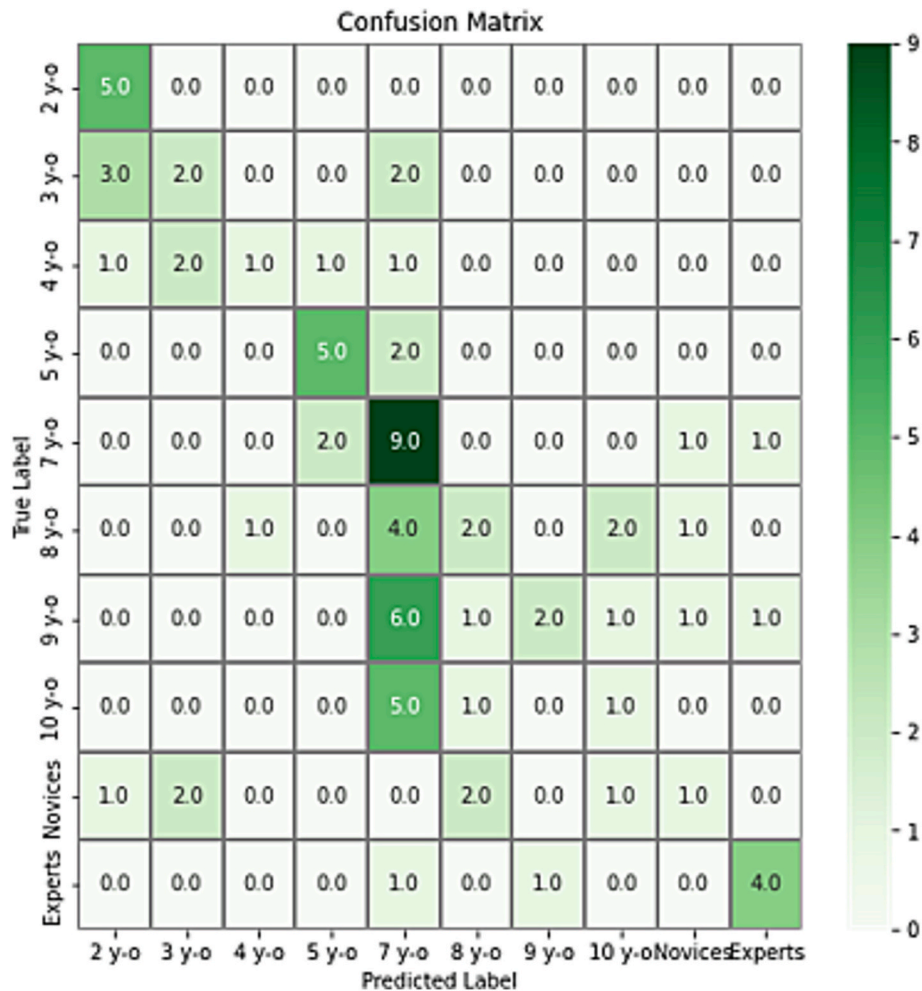


Fig. 2. Confusion matrix for the model classifying drawings into 10 distinct age categories. Each cell in the matrix indicates the number of drawings assigned by the model to a predicted category (shown in the columns) relative to their true category (shown in the rows). The diagonal cells, where the predicted category matches the true category, represent the correctly classified drawings for each age group and thus reflect the model's per-class accuracy. Off-diagonal cells display misclassifications, indicating which age groups are most frequently confused with others by the model. For example, a high value in an off-diagonal cell suggests systematic confusion between two specific age groups, while uniformly low off-diagonal values suggest better discrimination. This matrix offers a detailed overview of the model's performance by visualising both the successes and the error patterns across all categories, helping to identify which age groups are most distinct in their drawing characteristics and which present greater classification challenges due to developmental or stylistic overlap.

both RGB and grayscale models ($F_{6,126} = 38.41$, $p < 0.0001$), as depicted in Fig. 3. Notably, the grayscale models generally exhibit higher mean accuracy than their RGB counterparts ($F_{1,126} = 7.09$, $p = 0.009$), with the exception of the model for the 3–4 years old category. There is indeed an interaction effect between colours and age categories on the accuracy ($F_{6,126} = 5.54$, $p < 0.0001$). The models exhibit a strong capability to distinguish between the drawings of 2–3-year-old children and those of adults. However, this differentiation accuracy diminishes progressively as the age increases, eventually stabilising at a plateau for the drawings produced by children in the 5–7-year-old age group. This suggests that while the models are highly effective at identifying the distinctive characteristics of very young children's drawings compared to those of adults, the differences become less pronounced or harder to detect in the artwork of older children, particularly those aged 5 to 7 years. This plateau indicates a point where the models no longer significantly improve in distinguishing between the drawings of children in this age range and adults, reflecting a nuanced challenge in capturing the gradual development of drawing skills as children grow older. The observed variation in performance across iterations can be attributed to the random initialization of model-specific layers and the stochastic application of horizontal flips during data augmentation, both of which introduce variability in the training process.

3.2. Models by pair

Further analysis involved predicting the age categories between children and adults to compute the mean prediction for each model and class. Each probability output ranges from 0 (closer to the child side of that specific pair) to 1 (closer to adults), allowing us to read the mean score of any age group as a continuous maturity index along the child-to-adult trajectory. The outcomes of these predictions are illustrated in Fig. 4 for both RGB and grayscale models, along with an

example interpretation of the matrices. To remind, we employed probability score ranging from 0 to 1, where a score of 0 represents approximation to drawings made by younger children and a score of 1 corresponds to drawings more similar to those made by adults. On the other hand, a score near the midpoint of 0.5 reveals a level of ambiguity, showing the model's challenge in clearly separating the artworks by these age groups. In the prediction matrices for RGB images, the first row illustrates that all mean predictions surpass the 0.5 threshold, indicating that drawings made by children aged from 4 to 10 years more closely resemble those made by adults than those of the 2–3-year-old category. Notably, in this first row and subsequent ones, the mean prediction values for each specific age group rise in conjunction with the age of the predicted category, from younger children to adults, offering a quantitative measure of the development of drawing skills. For example, the mean prediction value for 4-year-olds is 0.67, suggesting their drawings are more adult-like than those of 2–3-year-olds, as 0.67 significantly exceeds 0.5. Conversely, in the model for 3–4-year-olds, the prediction for 5-year-olds is 0.54, barely distinguishable from a random classification since this value hovers near 0.5. Focusing on the transition from drawings by 2–3-year-olds to those by adults, stabilisation in mean prediction values is observed from ages 7 to 10 years, as opposed to the notable increase observed from 4 to 7 years. This might indicate a plateau in the evolution of drawing skills within this age range. For models analysing children from 4 to 5 years old and older, the prediction values approximate 0.5, highlighting challenges in differentiating drawings, which may suggest a convergence in drawing styles. The model focusing on 5–7-year-olds indicates stabilisation, with mean prediction values approaching 0.5. This suggests that drawings by children aged 8 to 10 years are perceived as similarly adult-like to those by 5–7-year-olds. In the final two models, which compare children's drawings to those of 7–8 and 8–9-year-olds, predictions fall below 0.5 for the 8–9 and 9–10-year-old categories, potentially indicating a

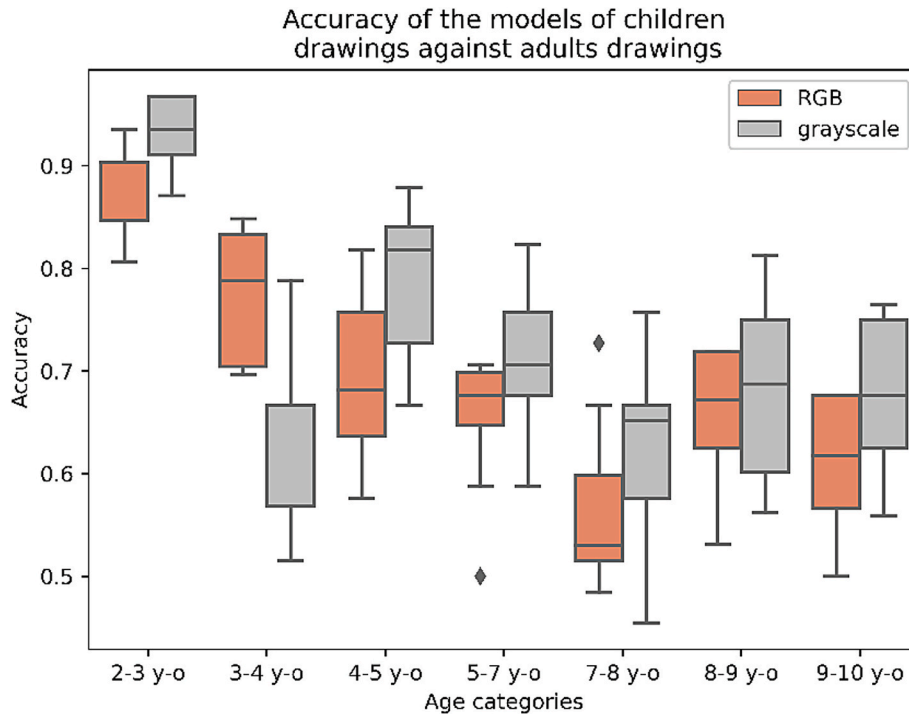


Fig. 3. Accuracy of the different binary classification models trained to distinguish between children's drawings and adults' drawings. Each data point in the graph represents the mean accuracy across 10 independent training iterations for a given child age group versus adults, ensuring that the reported values account for variability due to random model initialisation and data augmentation effects. The x-axis displays the child age groups used in each model, while the y-axis shows the corresponding mean classification accuracy. This figure provides a detailed view of how effectively the models can differentiate between children's and adults' drawings at each developmental stage, revealing trends such as high accuracy for distinguishing very young children's drawings from adults and a progressive decline in accuracy as the child age increases, reflecting the increasing similarity between older children's and adults' graphic productions. Overall, the figure allows readers to assess both the robustness of the models and how the discriminative visual features between age groups evolve over development.

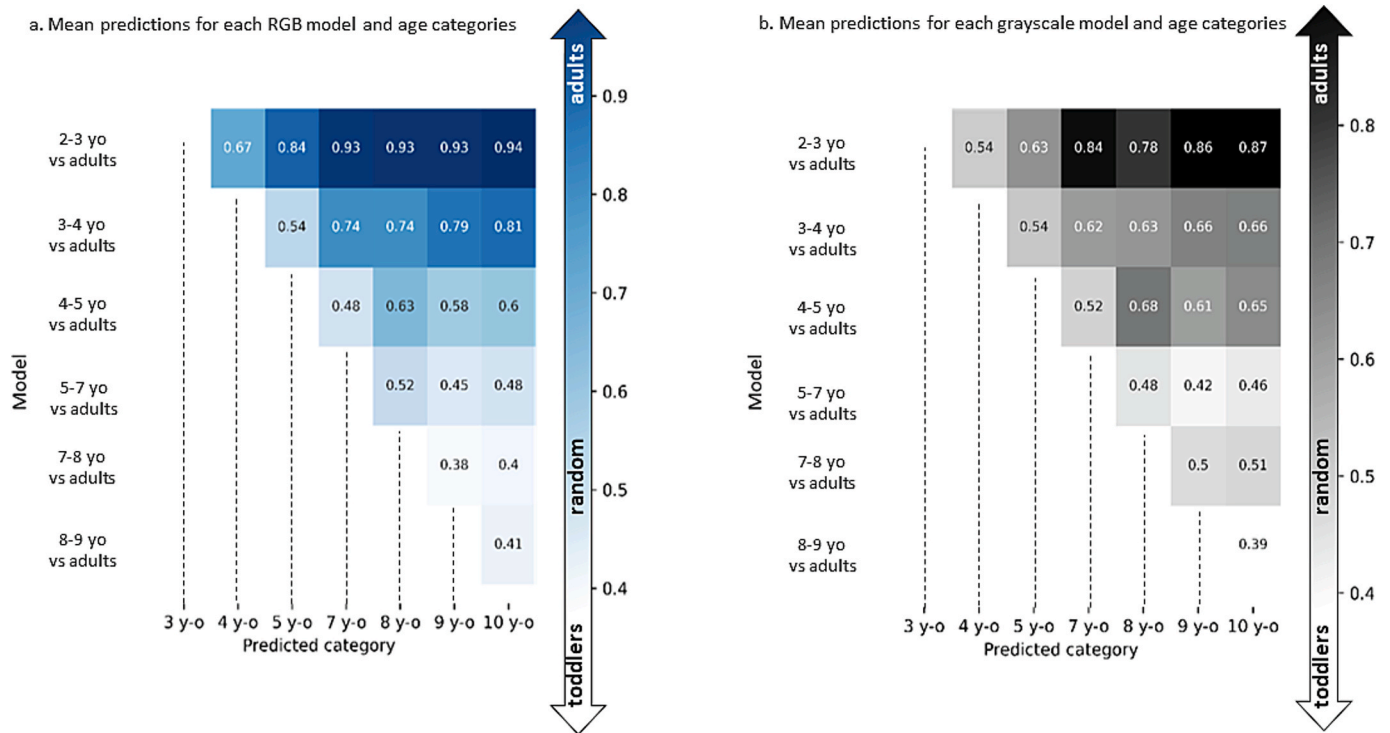


Fig. 4. Mean prediction values for each binary model across the different age categories, presented separately for (a) RGB models and (b) grayscale models. Each matrix shows, for every age group, the average predicted probability output by each model, reflecting the perceived similarity between the drawings of that age group and the two target classes (children vs. adults). The rows of the matrix correspond to the binary models trained on successive one-year child cohorts versus adults (e.g., 2–3 years vs. adults, 3–4 years vs. adults), while the columns indicate the age group of the drawings being evaluated, regardless of whether those drawings were part of the model's training. Each cell thus provides a quantifiable measure of how the model perceives the input drawings: values near 0 indicate that the drawings resemble the younger child category, and values near 1 suggest greater similarity to adult drawings. For example, in panel (a), the first row allows us to assess how the 2–3-year vs. adult model classifies drawings from 2 to 3-year-olds, 4–5-year-olds, or adults, offering insight into how discriminability changes across ages. A mean prediction of 0.84 for 5-year-old drawings means that, on average, the model considers them much closer to the adult reference than to the younger child group. This matrix-based approach enables a detailed and systematic understanding of developmental progression in drawing features, capturing both the gradual maturation process and the influence of colour information (RGB vs. grayscale) on model perception and classification outcomes.

slowdown in the development of drawing behaviour from ages 7 to 10. Based on these observations of prediction value increases, stabilisation, and their positions relative to the 0.5 uncertainty threshold, we can delineate three stages in drawing: 2 to 4 years old, about 4 to 7 years old, and above 7 years old.

3.3. Interpretability models

Grad-CAM and Captum, tools for model interpretability, were specifically applied to the validation datasets to investigate what features the VGG19 and ResNet18 convolutional neural networks (CNNs) identify and utilise to characterise and classify drawings. The interpretability of the Grad-CAM outputs varies significantly across different models and is also influenced by the representativeness of the drawings under examination, as illustrated in Fig. 5 and Fig. 6.

This variability is particularly evident when comparing the analysis of figurative drawings to that of non-figurative ones. In the case of figurative productions, it is relatively straightforward to discern what the model recognises as key features, such as faces or eyes. However, for non-figurative drawings, which lack clear representational content, it becomes challenging to understand what aspects of the drawing are being recognised and how they contribute to the classification decision. Furthermore, even when specific elements like rainbows or flowers are identified within a drawing, it remains unclear how the CNNs interpret these elements in the context of classifying the drawings.

A manual evaluation of the interpretability models, specifically analysing their performance through Grad-CAM heatmaps and Captum pixel attributions, revealed that 72 % of human faces in drawings were

accurately recognized as faces. This indicates that the models tend to focus more intensely on the areas depicting faces in drawings that contain them, as demonstrated by more pronounced heatmaps or pixel attributions in these regions. For instance, the mean number of positive pixels is $139,778 \pm 24,086$ for self-portraits whilst it is significantly lower and equals $102,352 \pm 35,893$ for free drawings (Wilcoxon rank sum test, $W = 17720$, $p < 0.0001$). However, this assessment is inherently subjective, reflecting the challenges in quantifying model interpretability. The models encountered difficulties in accurately identifying faces under certain conditions: when multiple faces appear within a single drawing, when faces are intertwined or merged with other lines or shapes, or when the drawings include animal faces. These challenges likely stem from the training dataset's composition, which predominantly featured self-portraits containing a single, clearly delineated human face. Moreover, in non-figurative or abstract drawings, the heatmaps seemed to lack coherence, with attention scattered across the image. This suggests that the models might rely more on abstract spatial patterns or contrasts rather than semantically meaningful features. Interestingly, the models sometimes focused on non-drawn areas, as observed in several Grad-CAM outputs (about 56 %). Since the drawings were created on tablet PCs, these non-drawn areas correspond strictly to blank regions without hidden colours or artefacts. This focus on blank regions could be attributed to their influence on spatial organization or contrast, which the models might interpret as distinguishing features.

Further analysis of the Grad-CAM and Captum outputs revealed additional challenges in the interpretability of the models, particularly in the context of figurative drawings. The models consistently prioritized facial features such as eyes and mouths of humans and non

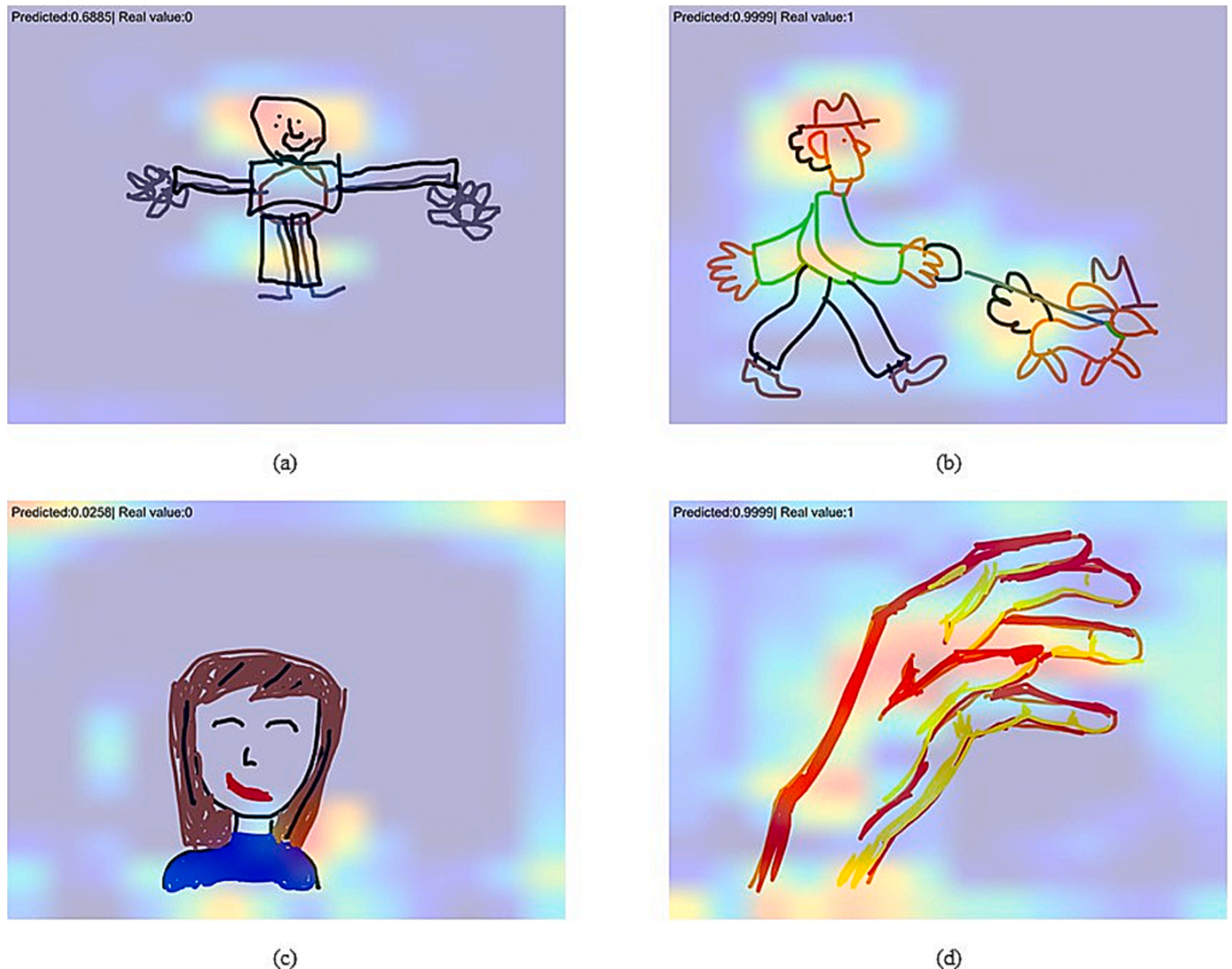


Fig. 5. This figure presents a detailed visual exploration of Grad-CAM (Gradient-weighted Class Activation Mapping) heatmaps generated for a selection of drawings from the validation sets of different binary classification models, showcasing how the models focus on specific regions of the images to make their predictions. Grad-CAM is a widely used technique in deep learning interpretability, designed to highlight the regions of an input image that contribute most to the decision-making process of a convolutional neural network (CNN). By overlaying colour-coded heatmaps onto the original images, Grad-CAM offers an intuitive visualisation of which parts of the drawing the model considers important when determining whether a drawing belongs to the child or adult category. Panel (a) features a heatmap overlay on a drawing by a 5-year-old child, analysed using the 4–5-year-old model. Interestingly, even though the drawing was misclassified as an adult's production, with a predicted probability of 0.6885, the heatmap highlights the facial region as the most salient feature, suggesting that the model relies heavily on the presence or quality of facial elements when making its age predictions. Panel (b) shows the heatmap for a drawing by an adult expert, evaluated with the 2–3-year-old model, where the model correctly classifies the drawing, and the heatmap displays a concentrated focus on specific, presumably informative, features of the image, contributing to the model's confident decision. Panel (c) presents a drawing by a 10-year-old child, assessed by the 9–10-year-old model. Although the model correctly classifies the drawing, the corresponding heatmap appears scattered or nonsensical, failing to highlight any clearly interpretable features, which raises questions about the transparency and consistency of the model's internal reasoning process. Panel (d) displays another adult expert drawing, again evaluated with the 2–3-year-old model, where, despite a high-confidence prediction, the heatmap fails to highlight any particular area of the drawing, suggesting that the model's decision-making process does not always align with human-intuitive understandings of feature importance. Taken together, these examples illustrate both the potential and the limitations of using Grad-CAM for model interpretability in the context of age classification of drawings. While some heatmaps align with human expectations—focusing on meaningful elements like faces or structured details—others produce puzzling or diffuse attention maps, underlining the challenges of interpreting the inner workings of deep neural networks. This figure therefore not only provides insight into what features the models rely on but also highlights the complexity and variability of model explanations, offering a valuable perspective on the current interpretability and reliability of deep learning approaches in developmental analysis of artistic production.

humans, demonstrating alignment with human intuition about the importance of these elements in visual representations. For instance, among 3-year-old children, facial elements appear in approximately 10 % of free drawings and 60 % of self-portraits. By the age of 7, the prevalence of faces increases significantly, reaching 65 % and 100 % in respectively free drawings and self-portraits. Among novice adults, facial elements are consistently present in 62 % and 100 % in respectively free drawings and self-portraits. This progression highlights a

developmental trend: as children age, not only does the frequency of faces in their drawings increase, but the complexity and detail of facial elements also improve. These include nuanced features such as the accurate depiction of eyes, mouths, and proportions. The models appear to leverage these developmental patterns to classify the age of the artist effectively. The recognition of such features, along with their increasing sophistication, suggests that the models are learning to associate detailed facial elements with specific age groups. However, the mean

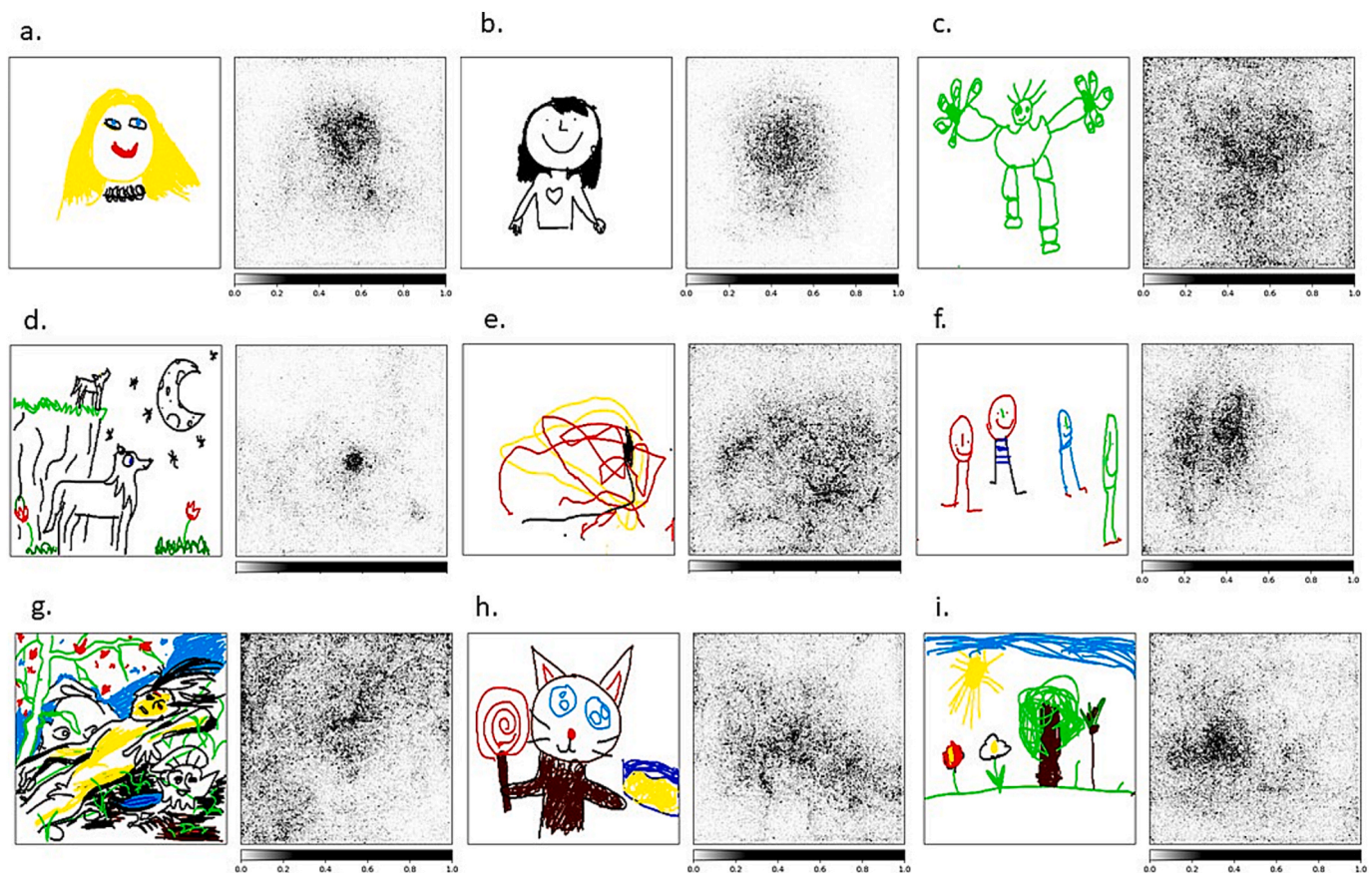


Fig. 6. This figure presents a series of examples using Captum, an interpretability tool designed to visualise which elements of an image are most influential in a model's classification decisions. Specifically, in these examples, black pixels are used to indicate the regions that the model deems most important for determining the age category of a drawing. Captum works by computing attribution maps that highlight the contribution of each pixel or region to the final prediction, helping researchers better understand the model's internal reasoning. Panel (a) shows an example where the model accurately identifies a human face as the most important feature in the drawing, focusing its attention on critical facial details that are likely informative for age-related classification. Panel (b) similarly presents another successful identification of a face, demonstrating the model's ability to consistently detect and prioritise key human features when available. Panel (c), however, illustrates a different pattern: here, the model attributes most of its attention to the body of a figure, while failing to highlight the face, suggesting that body shapes or postures can sometimes serve as alternative cues in the classification process, particularly when facial features are underdeveloped or absent in children's drawings. Panel (d) highlights the eye in a drawing of a wolf, indicating that the model is capable of focusing on non-human facial features when making decisions, especially in free-drawing tasks where animals or imagined characters are depicted. Panel (e) presents a scribble, a common type of production among younger children, where Captum's attribution map still selects certain areas as influential, although the meaning or logic behind these selections may not always be clear from a human perspective. Panel (f) shows a drawing with multiple human faces, of which only two are prominently identified by Captum, pointing to the model's partial success and its limitations when handling complex or crowded scenes. Panel (g) depicts an example where three faces appear fused or overlapped with other elements, creating a visual challenge that leads to incomplete or confused recognition by the model. Panel (h) shows a drawing of a cat where the model fails to recognise the cat's face as an important feature, possibly due to its training bias towards human forms, while panel (i) interestingly demonstrates a case where flowers—not faces or bodies—are marked as the most influential elements, suggesting that decorative or non-representational features can also play a role in shaping the model's predictions. Together, these examples offer a nuanced view of how the model distributes attention across diverse visual elements, revealing both its capacity for meaningful feature detection and the interpretability challenges that arise when applying deep learning methods to the analysis of children's and adults' drawings across varying styles, subjects, and levels of abstraction.

number of positive pixels is not different between age categories (Kruskal-Wallis test, $\chi^2 = 7.21$, $df = 8$, $p = 0.514$) indicating the importance of other parameters in discriminating drawings as other elements or blank space.

4. Discussion

Drawings have long been recognised as a reflective mirror to the inner workings of the human mind, especially in children. Yet, the interpretation of these drawings by adults may not always align with the child's original intent. This discrepancy underscores the potential for bias, particularly when interpretations rely on subjective judgment rather than objective or mathematical definitions [24,33,34]. In response to this challenge, we advocate for the adoption of novel and objective methodologies capable of decoding the rich tapestry of

information encapsulated within drawings [15]. Our research harnesses the power of deep learning to navigate the complex landscape of children's and adults' drawings, offering new insights into the evolution of drawing behaviour across different ages.

The initial model, tasked with categorising drawings into ten distinct age groups, achieved 40 % accuracy, far surpassing the 10 % expected from random guesswork. However, the preponderance of drawings classified by mistake as belonging to 7-year-olds suggests an anomaly likely attributed to the model's internal 'black box' mechanics rather than to any tangible psychological or developmental rationale. By adopting a paired comparison approach, we found that models distinguishing between the youngest children (2 and 3 years old) and adults were particularly effective, achieving accuracy rates exceeding 85 %. This accuracy diminishes with age, plateauing around 60 % for models comparing 7 and 8-year-olds to adults—a trend that intuitively mirrors

the increasing sophistication of children's drawings at this age. Intriguingly, converting images to grayscale improved accuracy across almost all models, with the notable exception of the 3–4-year-old category, where colour appears to play a pivotal role in the expressiveness of drawings, possibly serving as a tool for exploration rather than representation [24]. Our analysis further may identify three potential stages within the prediction matrices, representing the trajectory of drawing behaviour. This stage not only corroborate previous findings [4,7,35] but also may offer a potential framework for understanding the progression from scribbles to more sophisticated artistic expressions, aligning with established theories like those proposed by Luquet [3].

The interpretability of deep learning models [36,37], especially when examined through the lenses of Grad-CAM and Captum, presents a complex landscape that is as varied as it is intriguing. These tools, designed to provide a window into the 'thought processes' of neural networks, generate heatmaps or pixel graphs that can sometimes clearly demarcate the features deemed important by the model, such as human and non human faces in drawings. This capability is remarkable, suggesting that, to some extent, models are capable of 'seeing' and prioritising elements in images that humans also find significant. This endeavour could be significantly advanced by increasing the sample size and refining the models' ability to identify specific facial features such as eyes, nose, mouth, etc. [2,5,38]. As children grow, their drawings change in noticeable ways. Young children, around 2–3 years old, start with simple scribbles. At this age, they draw mainly for fun, enjoying the movement of their hands rather than trying to represent anything specific. Their drawings are quick, use very few sequences, and often look random to an adult's eye [35,39,40]. They also love experimenting with lots of colours, as they are more interested in playing than creating something meaningful. By the time children reach 7 or 8 years old, their drawings become much more organized [24,34]. They focus on showing all the parts of an object they want to draw, such as drawing a house with a roof, windows, and a door. At this stage, their goal is to make their drawings easy to understand, and they leave out unnecessary details [24]. They use fewer colours than younger children because their approach is more efficient—they care more about clarity than decoration. Their drawings reflect a growing ability to plan and control their movements. Research shows a consistent progression in children's drawings across cultures, beginning with scribbles, then simple figurative sketches, and eventually detailed representations [3,5,41] that we observed in our drawing dataset [24,34].

Enhancing the granularity with which the models recognise and interpret these elements could lead to a deeper understanding of the nuanced ways in which neural networks process visual information, offering more detailed insights into their interpretive capabilities. However, the clarity and utility of these heatmaps are not uniform across all types of drawings. When these interpretability tools are applied to non-figurative drawings that do not directly represent visible objects or scenes, their output often becomes cryptic. This enigmatic nature of the heatmaps in such contexts highlights a fundamental challenge in artificial intelligence: understanding how deep learning models process and interpret abstract visual content. Unlike figurative productions, where the presence of recognisable shapes and forms can guide the interpretation of heatmaps, nonfigurative ones lack these anchors, making the model's focus and decision-making process harder to decipher.

This variability in interpretability underscores a broader issue within the field of AI—despite the advanced capabilities of neural networks, their decision-making processes can sometimes be as opaque as they are sophisticated [13,42–44]. The challenge lies not only in achieving high accuracy in tasks such as drawing classification but also in making these processes transparent and understandable. This is particularly crucial when AI is used in domains where understanding the *why* behind decisions is as important as the decisions themselves. Moreover, even with we got respectable results to classify drawings according to age, the discrepancy in heatmap's clarity between figurative and non-figurative

drawings raises questions about the training of these models. Neural networks learn to prioritise certain features over others based on the datasets on which they are trained. If these datasets are predominantly composed of figurative images, the models may develop a bias towards recognising and interpreting features that are present in such images, at the expense of understanding more abstract, nonfigurative content. This suggests that diversifying training datasets to include a broader range of artistic expressions could be key to enhancing model interpretability across a wider spectrum of drawings.

Moreover, the focus of the models on non-painted areas, as observed in some instances using Grad-CAM and Captum (Fig. 5), reveals intriguing insights into how spatial organization and blank spaces may play a role in classification. This behaviour does not reflect an issue with the input data itself, as the drawings were created on tablet PCs with no hidden colours or background artefacts—white areas correspond strictly to non-drawn regions. Instead, it suggests that the models might be leveraging the spatial organization and blank spaces as features indicative of age-related progression in drawing styles. This is confirmed by the optimal model, which achieves an accuracy of 88 % in distinguishing self-portraits from free drawings, whereas it would be theoretically only 66.1 % if the model focused solely on faces. As children develop, their drawings evolve from scattered scribbles that fill the entire canvas to more organized compositions, where elements are deliberately placed, and blank spaces appear in specific regions [4,45]. For example, younger children often fill the page with unstructured marks, leaving little to no blank space. By contrast, older children and adults tend to centre key elements, such as houses or figures, in the middle of the page and reserve certain areas, like the corners, for context-specific features such as the sun or clouds. These blank spaces are not random [34] but are instead part of an intentional spatial organization that reflects the drawer's growing cognitive and motor skills. The deep learning models, therefore, might interpret these organizational patterns and the strategic use of blank spaces as critical features for classifying drawings and estimating age. The contrast between drawn and undrawn regions could highlight the increasing regularity and intentionality in spatial layout as children age, with well-centred drawings and proportionally arranged elements emerging as distinguishing characteristics. Additionally, convolutional neural networks analyse all regions of an image, including undrawn areas, during feature extraction. The sharp contrast between blank spaces and drawn regions might inadvertently draw the model's attention to these areas, even if their relevance is less apparent to human observers. This suggests that the models might be implicitly recognizing and utilizing the overall structure and balance of the drawing, including how blank spaces frame and contextualize the depicted elements. To improve interpretability and ensure the models prioritize semantically meaningful features, future refinements could focus on incorporating constraints or techniques that better disentangle relevant elements, such as spatial organization and intentional use of blank spaces, from potentially less significant areas. This could help reveal deeper insights into how these models process and classify artistic development in children's drawings.

Furthermore, the interpretability challenge also points to the need for developing more sophisticated tools and techniques that can provide deeper insights into the workings of neural networks. As the field of AI continues to evolve, the quest for models that are not only accurate but also interpretable will likely remain a central theme, driving advancements in technology and methodology. Moreover, factors such as the emotional state and motivation of the child, as well as the conditions under which the drawings were produced, can introduce variability into our analysis. Despite these challenges, ensuring uniform conditions across all age groups helps mitigate potential biases, suggesting that future research protocols could benefit from a more controlled drawing environment. To further refine our models and enhance their predictive accuracy, we propose expanding the scope of our trials and exploring alternative architectural frameworks. Additionally, comparing machine-generated classifications with human judgments could provide valuable

insights into the interpretability and applicability of these models in real-world contexts.

Drawing style which may be different according to countries will be interesting to implement in such artificial intelligence model. As all participants were French in this study, we could not test the different styles due to cultural differences. However, a previous study [2] analysed 958 self-portraits drawn by children aged 2–15 years from 35 countries across five continents, revealing cultural and age-related variations in complexity and the representation of social and physical environments, influenced by factors such as urbanization and individualism. Urban children depicted more detailed faces and symbols, individualistic societies featured larger self-representations and fewer man-made elements, and some cultural groups included more natural elements or personifying their environment, showcasing the influence of sociocultural contexts on graphical expression. Deep learning models could be able to extract such features and classify drawings according to cultures.

In conclusion, our research highlights how deep learning models can classify, even with some difficulties, drawings across age groups. Work still has to be done but this is an important methodological step in our understanding of drawing behaviour. Indeed, AI and more particularly deep learning can now be considered as a new tool in our pre-existing drawing comprehension ‘tool box’ including other devices as fractals, PCA, etc. [15]. This allows new perspectives of interdisciplinary work and underscores the potential of deep learning to uncover the subtle nuances of human expression and perception. Indeed, such approach could revolutionise in the diagnosis of mental health conditions, such as depression [46,47] or developmental disorders [48,49], by providing nuanced insights into patients’ mental states through their drawings. Tasks like the Clock Drawing Test (CDT) have shown promise in identifying cognitive impairments such as Alzheimer’s disease, where features like spatial organization and stroke pressure reveal subtle [50,51]. Similarly, unique traits in drawings by children with autism spectrum disorder (ASD), combined with eye-tracking data, could enhance early diagnostic tools [52,53]. Also, our findings offer a new lens to understand how human cultures and societies influence drawing behaviour and vice versa [1,2,54]. Eventually, deep learning could enrich discussions on how drawing – and by extension art – may reflect societal values and experiences. Expanding datasets to include diverse cultural contexts would also uncover cross-cultural differences in drawing behaviour, enriching models’ interpretability and generalizability [55]. Beyond diagnostics, these models can support personalized educational tools, assessing developmental milestones and offering targeted feedback. Art therapy programs could leverage AI-based drawing analysis to monitor therapeutic progress, improving quality of life for individuals with cognitive impairments [56]. Interdisciplinary research combining art, psychology, and AI could further explore how drawing reflects cultural and emotional states, advancing both scientific understanding and practical applications in diagnostics and therapy. We hope our study paves the way for future explorations into the multidimensional expression that is drawing.

CRediT authorship contribution statement

Benjamin Beltzung: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Marie Pelé:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. **Lison Martinet:** Data curation, Validation, Writing – review & editing. **Elliot Maitre:** Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – review & editing. **Jimmy Falck:** Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – review & editing. **Cédric Sueur:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration,

Supervision, Validation, Writing – original draft, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.displa.2025.103166>.

Data availability

Codes and scripts are available on <https://github.com/cedricsueur/drawinganalyses>. Data is available on Zenodo: [Doi: 10.5281/zenodo.11097174](https://doi.org/10.5281/zenodo.11097174).

References

- [1] B. Binson, D.J. Federman, R. Lev-Wiesel, Do self-figure drawings reveal the drawer’s cultural values? Thais and Israelis draw themselves, *J. Humanist. Psychol.* (2019) 0022167819831082, <https://doi.org/10.1177/0022167819831082>.
- [2] S. Restoy, L. Martinet, C. Sueur, M. Pelé, Draw yourself: how culture influences drawings by children between the ages of two and fifteen, *Front. Psychol.* 13 (2022) 940617.
- [3] G.-H. Luquet, *Le dessin enfantin*. (Bibliothèque de psychologie de l’enfant et de pédagogie.), (1927).
- [4] E. Adi-Japha, I. Levin, S. Solomon, Emergence of representation in drawing: the relation between kinematic and referential aspects, *Cogn. Dev.* 13 (1998) 25–51.
- [5] R. Baldy, Dessin et développement cognitif, *Enfance* 57 (2005) 34–44.
- [6] M.V. Cox, *Children’s drawings of the human figure*, Psychology Press, 2013.
- [7] N.H. Freeman, *Drawing: Public instruments of representation*, (1993).
- [8] S. Khan, T. Yairi, A review on the application of deep learning in system health management, *Mech. Syst. Sig. Process.* 107 (2018) 241–265, <https://doi.org/10.1016/j.ymssp.2017.11.024>.
- [9] O.M. Parkhi, A. Vedaldi, A. Zisserman, Deep face recognition. *Proceedings of the British Machine Vision Conference 2015*, British Machine Vision Association, Swansea, 2015, p. 41.1–41.12. [Doi: 10.5244/C.29.41](https://doi.org/10.5244/C.29.41).
- [10] M.D. Hassan, A.N. Nasret, M.R. Baker, Z.S. Mahmood, Enhancement automatic speech recognition by deep neural networks, *Periodicals of Engineering and Natural, Sciences* 9 (2021) 921–927, <https://doi.org/10.21533/pen.v9i4.2450>.
- [11] R. Chauhan, K.K. Ghanshala, R. Joshi, Convolutional neural network (CNN) for image detection and recognition, *IEEE* (2018) 278–282.
- [12] T. Lei, Z. Shi, D. Liu, L. Yang, F. Zhu, A novel CNN-based method for question classification in intelligent question answering, in: 2018: pp. 1–6.
- [13] R. Shwartz-Ziv, N. Tishby, Opening the black box of deep neural networks via information, *arXiv Preprint arXiv:1703.00810* (2017).
- [14] B. Beltzung, M. Pelé, J.P. Renoult, M. Shimada, C. Sueur, Using Artificial intelligence to analyze non-human drawings: a first step with orangutan productions, *Animals* 12 (2022) 2761, <https://doi.org/10.3390/ani12202761>.
- [15] B. Beltzung, M. Pelé, J.P. Renoult, C. Sueur, Deep learning for studying drawing behavior: a review, *Front. Psychol.* 14 (2023) 992541, <https://doi.org/10.3389/fpsyg.2023.992541>.
- [16] S.-Y. Chen, P.-H. Lin, W.-C. Chien, Children’s digital art ability training system based on AI-assisted learning: a case study of drawing color perception, *Front. Psychol.* 13 (2022) 823078, <https://doi.org/10.3389/fpsyg.2022.823078>.

- [17] A. Philippsen, Y. Nagai, A predictive coding account for cognition in human children and chimpanzees: a case study of drawing, *IEEE Trans. Cogn. Dev. Syst.* (2020) 1, <https://doi.org/10.1109/TCDS.2020.3006497>.
- [18] N. Ali, A. Abd-Alrazaq, Z. Shah, M. Alajlani, T. Alam, M. Househ, Artificial intelligence-based mobile application for sensing children emotion through drawings, *Stud. Health Technol. Inform.* 295 (2022) 118–121.
- [19] L. Kissos, L. Goldner, M. Butman, N. Eliyahu, R. Lev-Wiesel, Can artificial intelligence achieve human-level performance? A pilot study of childhood sexual abuse detection in self-figure drawings, *Child Abuse Neglect* 109 (2020) 104755, <https://doi.org/10.1016/j.chiabu.2020.104755>.
- [20] B. Long, J.E. Fan, M.C. Frank, Drawings as a window into developmental changes in object representations. *Proceedings of the 40th Annual Conference of the Cognitive Science Society*, 2018.
- [21] J. Moon, M.-J. Kim, S.-O. Lee, Y. Yu, A deep learning model based on triplet losses for a similar child drawing selection algorithm, *J. Korea Indust. Inf. Syst. Res.* 27 (2022) 1–9, <https://doi.org/10.9723/jksis.2022.27.1.001>.
- [22] D. Pysal, S.J. Abdulkadir, S.R.M. Shukri, H. Alhussian, Classification of children's drawing strategies on touch-screen of seriation objects using a novel deep learning hybrid model, *Alex. Eng. J.* 60 (2020) 115–129.
- [23] Y. Yuan, J. Huang, X. Ma, K. Yan, Children's drawing psychological analysis using shallow convolutional neural network, in: *2020 International Conferences on Internet Of Things (IThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData) and IEEE Congress on Cybermatics (Cybermatics)*, IEEE, 2020, pp. 692–698.
- [24] L. Martinet, C. Sueur, S. Hirata, J. Hosselet, T. Matsuzawa, M. Pelé, New indices to characterize drawing behavior in humans (*Homo sapiens*) and chimpanzees (*Pan troglodytes*), *Sci. Rep.* 11 (2021) 3860, <https://doi.org/10.1038/s41598-021-83043-0>.
- [25] L. Torrey, J. Shavlik, Transfer learning, in: *Handbook of Research on Machine Learning Applications and Trends: Algorithms, Methods, and Techniques*, IGI Global, 2010, pp. 242–264. [Doi: 10.4018/978-1-60566-766-9.ch011](https://doi.org/10.4018/978-1-60566-766-9.ch011).
- [26] K. Simonyan, A. Zisserman, Very Deep Convolutional Networks for Large-Scale Image Recognition, *arXiv:1409.1556 [Cs]* (2015). <http://arxiv.org/abs/1409.1556> (accessed November 10, 2021).
- [27] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, F.-F. Li, Imagenet: A large-scale hierarchical image database, in: *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255, [10.1109/CVPR.2009.5206848](https://doi.org/10.1109/CVPR.2009.5206848).
- [28] A. Theodorus, M. Nauta, C. Seifert, Evaluating CNN interpretability on sketch classification, in: *Twelfth International Conference on Machine Vision (ICMV 2019)*, 2020, pp. 475–482, [10.1117/12.2559536](https://doi.org/10.1117/12.2559536).
- [29] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, *In* (2016:) 770–778.
- [30] C. Guo, G. Pleiss, Y. Sun, K.Q. Weinberger, On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1321–1330.
- [31] R.R. Selvaraju, A. Das, R. Vedantam, M. Cogswell, D. Parikh, D. Batra, Grad-CAM: Why did you say that?, *arXiv:1611.07450 [Cs, Stat]* (2017). <http://arxiv.org/abs/1611.07450> (accessed April 26, 2022).
- [32] N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, Captum: A unified and generic model interpretability library for pytorch, *arXiv Preprint arXiv:2009.07896* (2020).
- [33] B. Beltzung, L. Martinet, A.J. Macintosh, X. Meyer, J. Hosselet, M. Pelé, C. Sueur, To draw or not to draw: understanding the temporal organization of drawing behavior using fractal analyses, *Fractals* 31 (2023) 2350009.
- [34] C. Sueur, L. Martinet, B. Beltzung, M. Pelé, Making drawings speak through mathematical metrics, *Hum. Nat.* 33 (2022) 400–424.
- [35] J. Matthews, Children drawing: are young children really scribbling? *Early Child Dev. Care* 18 (1984) 1–39, <https://doi.org/10.1080/0300443840180101>.
- [36] A. Schöttl, A light-weight method to foster the (Grad) CAM interpretability and explainability of classification networks, *IEEE* (2020) 348–351.
- [37] Q. Zhang, S.-C. Zhu, Visual interpretability for deep learning: a survey, *Front. Inf. Technol. Electron. Eng.* 19 (2018) 27–39.
- [38] R. Baldy, Fais-moi un beau dessin: regarder le dessin de l'enfant, comprendre son évolution, *In Press*, 2011.
- [39] H. Gardner, Artful scribbles: The significance of children's drawings, (1981).
- [40] C. Longobardi, R. Quaglia, N.O. Iotti, Reconsidering the scribbling stage of drawing: a new perspective on toddlers' representational processes, *Front. Psychol.* 6 (2015) 1227.
- [41] C. Golomb, *The Child's Creation of a Pictorial World*, Psychology Press, 2003.
- [42] M. Carabantes, Black-box artificial intelligence: an epistemological and critical analysis, *AI & Soc.* 35 (2020) 309–317.
- [43] E. Duede, Deep learning opacity in scientific discovery, *Philos. Sci.* 90 (2023) 1089–1099.
- [44] F. Faries, V. Raja, Black Boxes and Theory Deserts: Deep Networks and Epistemic Opacity in the Cognitive Sciences, (2022).
- [45] R. Kellogg, *Analyzing Children's Art, Social Sciences & World Languages*, McGraw-Hill Humanities, 1969.
- [46] L. Eytan, D.L. Elkins-Abuhoff, Indicators of depression and self-efficacy in the PPAT drawings of normative adults, *Arts Psychother.* 40 (2013) 291–297, <https://doi.org/10.1016/j.aip.2013.04.003>.
- [47] J. Kim, S. Chung, Drawing test form for depression: the development of drawing tests for predicting depression among breast cancer patients, *Psychiatry Investig.* 18 (2021) 879–888, <https://doi.org/10.30773/pi.2021.0044>.
- [48] T. Charman, S. Baron-Cohen, Drawing development in autism: the intellectual to visual realism shift, *Br. J. Dev. Psychol.* 11 (1993) 171–185, <https://doi.org/10.1111/j.2044-835X.1993.tb00596.x>.
- [49] A. Lee, R.P. Hobson, Drawing self and others: how do children with autism differ from those with learning difficulties? *Br. J. Dev. Psychol.* 24 (2006) 547–565, <https://doi.org/10.1348/026151005X49881>.
- [50] J.Y.C. Chan, B.K.K. Bat, A. Wong, T.K. Chan, Z. Huo, B.H.K. Yip, T.C.Y. Kowk, K.K. F. Tsoi, Evaluation of digital drawing tests and paper-and-pencil drawing tests for the screening of mild cognitive impairment and dementia: a systematic review and meta-analysis of diagnostic studies, *Neuropsychol. Rev.* 32 (2022) 566–576, <https://doi.org/10.1007/s11065-021-09523-2>.
- [51] R. Binaco, N. Calzaretto, J. Epifano, S. McGuire, M. Umer, S. Emrani, V. Wasserman, D.J. Libon, R. Polikar, Machine learning analysis of digital clock drawing test performance for differential classification of mild cognitive impairment subtypes versus Alzheimer's disease, *J. Int. Neuropsychol. Soc.* 26 (2020) 690–700, <https://doi.org/10.1017/S1355617720000144>.
- [52] F. Shi, W. Sun, H. Duan, X. Liu, M. Hu, W. Wang, G. Zhai, Drawing reveals hallmarks of children with autism, *Displays* 67 (2021) 102000, <https://doi.org/10.1016/j.displa.2021.102000>.
- [53] Q. Wei, H. Cao, Y. Shi, X. Xu, T. Li, Machine learning based on eye-tracking data to identify autism spectrum disorder: a systematic review and meta-analysis, *J. Biomed. Inform.* 137 (2023) 104254, <https://doi.org/10.1016/j.jbi.2022.104254>.
- [54] P. Bozzato, C. Longobardi, Cross-cultural evaluation of children's drawings of gender role stereotypes in Italian and Cambodian students, *J. Psychol. Educ. Res.* 29 (2021) 97–115.
- [55] G. Maestri, A. Nicotra, S. Pomati, M. Canevelli, L. Pantoni, I. Cova, Cultural influence on clock drawing test: a systematic review, *J. Int. Neuropsychol. Soc.* 29 (2023) 704–714, <https://doi.org/10.1017/S1355617722000662>.
- [56] S.Y.M. Emblad, E.B. Mukaetova-Ladinska, Creative art therapy as a non-pharmacological intervention for dementia: a systematic review, *J. Alzheimer's Dis. Rep.* 5 (2021) 353–364, <https://doi.org/10.3233/ADR-201002>.